COUNITS OF COMPOSITE ADJUNCTIONS AND LEFT ADJOINTS OF COMPOSITE FUNCTORS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3811-3836, following the statement at 3792-3809.

Editorial note. Both conclusion items (existence of the left adjoint and the counit formula) are delivered by the same verification.

Allowed standard prerequisites (editorial).

- Adjunctions and their counits. Definition definition-adjoint of this chapter and the discussion following it, used at 3813-3817. The counits of (u,v) and (u',v') and the Hom-set bijection characterization used in the verification.

Source reference. Original definition 0037: sources/stacks-categories/pinned-original-source.tex, source lines 3561–3561. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ be categories. Let $v : \mathcal{A} \to \mathcal{B}$ and $v' : \mathcal{B} \to \mathcal{C}$ be functors with left adjoints u and u' respectively. Then*

(1) *The functor $v'' = v' \circ v$ has a left adjoint equal to $u'' = u \circ u'$.*
(2) *Given X in $\mathcal{A}$ we have*

$$\epsilon^v_X \circ u(\epsilon^{v'}_{v(X)}) = \epsilon^{v''}_X : u''(v''(X)) \to X \tag{0.1.1}$$

Where ϵ is the counit of the adjunctions.

Proof. Let us unwind the formula in (2) because this will also immediately prove (1). First, the counit of the adjunctions for the pairs (u, v) and (u', v') are maps $\epsilon^v_X : u(v(X)) \to X$ and $\epsilon^{v'}_Y : u'(v'(Y)) \to Y$, see discussion following Definition 0037. With u'' and v'' as in (1) we unwind everything

$$u''(v''(X)) = u(u'(v'(v(X)))) \xrightarrow{u(\epsilon^{v'}_{v(X)})} u(v(X)) \xrightarrow{\epsilon^v_X} X$$

to get the map on the left hand side of (0.1.1). Let us denote this by $\epsilon^{v''}_X$ for now. To see that this is the counit of an adjoint pair (u'', v'') we have to show that given Z in $\mathcal{C}$ the rule that sends a morphism $\beta : Z \to v''(X)$ to $\alpha = \epsilon^{v''}_X \circ u''(\beta) : u''(Z) \to X$ is a bijection on sets of morphisms. This is true because, this is the composition of the rule sending β to $\epsilon^{v'}_{v(X)} \circ u'(\beta)$ which is a bijection by assumption on (u', v') and then sending this to $\epsilon^v_X \circ u(\epsilon^{v'}_{v(X)} \circ u'(\beta))$ which is a bijection by assumption on (u, v). □

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